

Checkerboard

Sample run

Enter a positive integer: 5

```
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
```

Code

```
n=int(input("Enter a positive integer: "))
print()
for row in range(1,n+1):
    for column in range(1,n+1):
        if (row+column)%2==0:
            print("0",end=" ")
        else:
            print("1",end=" ")
    print()
```

Hollow Square

Sample run

Enter a positive integer: 5

```
*****
*   *
*   *
*   *
*   *
*****
```

Code

```
n=int(input("Enter a positive integer: "))
print()
for row in range(1,n+1):
    for column in range(1,n+1):
        if row==1 or row==n or column==1 or column==n:
            print("*",end="")
        else:
            print(" ",end="")
    print()
```

0/1 Triangle

Sample run

Enter a positive integer: 5

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Code

```
n=int(input("Enter a positive integer: "))
print()
for row in range(1,n+1):
    for column in range(1,row+1):
        if (row+column)%2==0:
            print("1",end=" ")
        else:
            print("0",end=" ")
    print()
```

X Pattern

Sample run

Enter a positive integer: 5

```
*      *
 *    *
  *
 *  *
*      *
```

Code

```
n=int(input("Enter a positive integer: "))
print()
for row in range(1,n+1):
    for column in range(1,n+1):
        if row==column or row+column==n+1:
            print("*",end="")
        else:
            print(" ",end="")
    print()
```